

3 METHOD

3.1 PROBLEM FORMULATION

Let $\mathcal{R}$ and $\mathcal{E}$ denote reaction and enzyme sets. Observed associations form a many-to-many bipartite graph $\mathcal{D} \subseteq \mathcal{R} \times \mathcal{E}$. We learn enzyme–reaction compatibility by independently mapping both modalities to a shared space. The complete reaction and enzyme encoders f_R, f_E give normalized outputs $\mathbf{z}_r = f_R(r)$ and $\mathbf{z}_e = f_E(e)$, yielding the cosine score $S(r, e) = \mathbf{z}_r^\top \mathbf{z}_e \in [-1, 1]$.

The same score supports $E \rightarrow R$ retrieval over reaction candidates $\mathcal{C}_R(e) \subseteq \mathcal{R}$ and $R \rightarrow E$ retrieval over enzyme candidates $\mathcal{C}_E(r) \subseteq \mathcal{E}$. Relevant candidates are each query’s observed neighbors in the test graph, allowing multiple correct results. Query sets, candidate pools, and positive neighborhoods can differ between directions. Learning uses only the training graph $\mathcal{D}_{\text{tr}} \subseteq \mathcal{R}_{\text{tr}} \times \mathcal{E}_{\text{tr}}$.

V4 first aligns the encoders, then refines their embeddings over the complete training graph before adding training–reaction coordinates.

3.2 ENZYME REPRESENTATION

Catalytic function depends on both sequence context and functional residues. We capture these signals using frozen ProtT5 residue embeddings $\mathbf{H}_e \in \mathbb{R}^{L \times 1024}$ (Elnaggar et al., 2022) and scores from a frozen SLEEC model (Gong et al., 2026), complemented by K learned residue summaries. The global ProtT5 feature is $\mathbf{x}_e^P = L^{-1} \sum_{i \in \mathcal{V}_e} \mathbf{h}_i$, where $\mathcal{V}_e$ indexes the L valid residues.

SLEEC-guided pooling. We suppress e on residue vectors and attention weights. SLEEC assigns each $\mathbf{h}_i \in \mathbb{R}^{1024}$ a logit ℓ_i , centered and clipped as $\tilde{\ell}_i = \text{clip}(\ell_i - L^{-1} \sum_{j \in \mathcal{V}_e} \ell_j, -10, 10)$. The SLEEC feature $\mathbf{x}_e^S \in \mathbb{R}^{1024}$ summarizes enzyme e by averaging its residue embeddings with normalized SLEEC weights:

$$a_i^S = (1 - \eta) \frac{\exp(\tilde{\ell}_i)}{\sum_{j \in \mathcal{V}_e} \exp(\tilde{\ell}_j)} + \frac{\eta}{L}, \quad \mathbf{x}_e^S = \sum_{i \in \mathcal{V}_e} a_i^S \mathbf{h}_i, \quad (1)$$

Here $\eta = 0.05$ retains a uniform contribution from every valid residue.

Learned residue views. We learn queries $\mathbf{q}_k \in \mathbb{R}^{256}$, $k = 1, \dots, K$, shared across enzymes. A shared residue adapter and separate key and value projections produce normalized keys $\mathbf{k}_i$ and unnormalized values $\mathbf{v}_i$ in $\mathbb{R}^{256}$. Write $\mathbf{n}(\mathbf{x}) = \mathbf{x} / \max(\|\mathbf{x}\|_2, \epsilon)$ for numerical normalization, with $\epsilon = 10^{-6}$ in this encoder. Query $\hat{\mathbf{q}}_k = \mathbf{n}(\mathbf{q}_k)$ defines

$$p_{ki} = \frac{\exp(\gamma \hat{\mathbf{q}}_k^\top \mathbf{k}_i)}{\sum_{j \in \mathcal{V}_e} \exp(\gamma \hat{\mathbf{q}}_k^\top \mathbf{k}_j)}, \quad \mathbf{x}_e^k = \sum_{i \in \mathcal{V}_e} \left[(1 - \eta) p_{ki} + \frac{\eta}{L} \right] \mathbf{v}_i, \quad (2)$$

Thus p_{ki} is a scalar weight over residues for query k ; the learned scale $\gamma = 10 \text{ sigmoid}(\rho_\gamma) \in (0, 10)$ controls concentration. Each view combines 95% attention-weighted and 5% uniform pooling of the values. Phase 1 trains the queries, adapter, and projections; frozen views can be cached per enzyme for reuse across candidate reactions. We retain $K = 4$ as a compromise across screening and retrieval: the single-seed $K \in \{1, 2, 4, 8\}$ ablation finds no uniformly best count (Appendix J).

Feature-wise gated fusion. We combine the extracted enzyme features $\{\mathbf{x}_e^P, \mathbf{x}_e^S\} \cup \{\mathbf{x}_e^k\}_{k=1}^K$ by projecting each feature into a common 256-dimensional space using an independent MLP followed by ℓ_2 normalization. The projected features are $\mathbf{y}_e^v = \mathbf{n}(F_v(\mathbf{x}_e^v))$, with view labels $v \in \mathcal{I} = \{P, S, 1, \dots, K\}$. A gating network maps their concatenation to feature-specific weights $\omega_{e, vd}$, normalized across views for each coordinate d . The fused representation is

$$\bar{\mathbf{y}}_{e, d} = \sum_{v \in \mathcal{I}} \omega_{e, vd} \mathbf{y}_{e, d}^v, \quad d = 1, \dots, 256. \quad (3)$$

Independent projections of $\mathbf{x}_e^P$ and $\bar{\mathbf{y}}_e$ yield normalized 512-dimensional vectors $\mathbf{g}_e$ and $\mathbf{f}_e$. The enzyme embedding is $\mathbf{b}_e = \mathbf{n}(\mathbf{g}_e + s \mathbf{f}_e)$, where the learned scalar $s \in (0, 0.5)$ controls the fused contribution (Appendix A.2).

3.3 REACTION ENCODER

The reaction encoder combines frozen ReactionT5v2, Uni-Mol2, and ChIRo features (Sagawa & Kojima, 2023; Ji et al., 2024; Adams et al., 2022) with a 617-dimensional chemistry descriptor computed from participant SMILES. This descriptor concatenates averaged Morgan fingerprints (512 coordinates), four aggregates of 16 molecular properties (64), participant-set statistics (4), and cofactor indicators (37). It supplies explicit substructure, physicochemical, and cofactor cues alongside pretrained embeddings. "Fixed" means its construction is not optimized during retrieval training; property normalization and the cofactor vocabulary are fitted on training reactions and frozen; its neural projection is learned (Appendix D).

For each molecular modality $m \in \{\text{UniMol2}, \text{ChIRo}\}$, independent attention poolers summarize reactant and product embeddings as weighted means $\mathbf{p}_-^m$ and $\mathbf{p}_+^m$. Their concatenation $\mathbf{c}_r^m = [\mathbf{p}_-^m; \mathbf{p}_+^m; \mathbf{p}_+^m - \mathbf{p}_-^m; |\mathbf{p}_+^m - \mathbf{p}_-^m|]$ retains both summaries alongside their signed and absolute differences. Independent MLPs map these two compositions, the ReactionT5v2 embedding, and the chemistry descriptor to four 512-dimensional tokens. Modality attention averages available tokens with learned weights; an output MLP and ℓ_2 normalization yield $\mathbf{b}_r$.

EnzymeMap supplies physical reactant and product sides. ReactZyme supplies an unordered participant set, copied to both poolers. Their parameters differ, so $\mathbf{p}_-^m$ and $\mathbf{p}_+^m$ can differ without representing an observed reaction direction. The chemistry descriptor combines participants irrespective of side. Appendix A.3 specifies the encoder.

3.4 ANCHOR-BALANCED MULTI-POSITIVE ALIGNMENT

We first align the encoders with a decoupled multi-positive objective, accounting for the many-to-many structure of enzyme–reaction associations. For anchor a , $P_B(a)$ contains all known training neighbors among the batch’s distinct opposite-modality endpoints; $N_B(a)$ contains the remainder.

With $s_{ax}^{(1)} = \mathbf{b}_a^\top \mathbf{b}_x / \tau_1$ and $\tau_1 = 0.2$, we minimize

$$\mathcal{L}_{\text{align}}^{(1)} = \frac{1}{2} \sum_{X \in \{R, E\}} \frac{1}{|\mathcal{A}_X|} \sum_{a \in \mathcal{A}_X} \frac{1}{|P_B(a)|} \sum_{p \in P_B(a)} \log \left(1 + \sum_{n \in N_B(a)} e^{s_{an}^{(1)} - s_{ap}^{(1)}} \right), \quad (4)$$

where $\mathcal{A}_X$ contains type- X anchors with nonempty positive and negative sets. Each positive competes only with $N_B(a)$, excluding other known positives from its denominator. Normalizing by $|P_B(a)|$ prevents anchors with more positives from receiving greater weight; the two directional losses are averaged equally. The implemented phase-1 objective also includes entropy and diversity penalties on residue attention (Appendix A.4).

3.5 FULL-GRAPH REFINEMENT

After initial alignment, we freeze both encoders and refine their representations over the complete training association graph. Two residual heads, h_E and h_R , learn additive adjustments to enzyme and reaction embeddings. For endpoint a of modality $X \in \{R, E\}$, the refined representation is $\mathbf{u}_a^{\text{tr}} = \mathbf{n}(\mathbf{b}_a + \kappa h_X(\mathbf{b}_a))$, where $\kappa = 0.2$ controls the residual contribution. We optimize the heads using cross-entropy with a uniform target distribution over each query’s known positive partners:

$$\mathcal{L}_{\text{CE}}^{(2)} = \frac{1}{2} \sum_{X \in \{R, E\}} \frac{1}{|\mathcal{T}_X|} \sum_{a \in \mathcal{T}_X} \left[\log \sum_{x \in \mathcal{T}_{\bar{X}}} e^{s_{ax}^{(2)}} - \frac{1}{|P(a)|} \sum_{p \in P(a)} s_{ap}^{(2)} \right], \quad (5)$$

where $\mathcal{T}_R = \mathcal{R}_{\text{tr}}$, $\mathcal{T}_E = \mathcal{E}_{\text{tr}}$, $\bar{X}$ denotes the opposite modality, and $P(a)$ contains the training positives of a . Scores are $s_{ax}^{(2)} = (\mathbf{u}_a^{\text{tr}})^\top \mathbf{u}_x^{\text{tr}} / \tau_2$, with $\tau_2 = 0.2$. Each query is compared with every opposite-modality training endpoint, including its positive partners. An identity penalty $\mathcal{L}_{\text{id}}$ discourages cosine displacement from the base embeddings, averaged equally over reactions and enzymes. The complete refinement objective is

$$\mathcal{L}^{(2)} = \mathcal{L}_{\text{CE}}^{(2)} + 10\mathcal{L}_{\text{id}}. \quad (6)$$

Only the residual heads are updated; both endpoint representations remain independently computable. Appendix A.5 gives the head architecture and identity-penalty definition.

3.6 BIDIRECTIONAL SCORING

A dictionary indexes training reactions. Frozen features are normalized, centered on training endpoints, and renormalized. Reaction coordinates $\mathbf{a}_r$ are normalized exponential cosine similarities to training reactions. For each enzyme, we find the 32 nearest training proteins; each reaction coordinate takes the maximum exponential similarity among linked neighbors. Normalization gives $\mathbf{a}_e$. Both kernels use temperature 0.03 (Appendix A.7).

For retrieval, we increase the fused protein contribution to $\tilde{\mathbf{b}}_e = \mathbf{n}(\mathbf{g}_e + 3s\mathbf{f}_e)$ and halve the residual coefficient, obtaining $\mathbf{u}_e = \mathbf{n}(\tilde{\mathbf{b}}_e + 0.1h_E(\tilde{\mathbf{b}}_e))$ and $\mathbf{u}_r = \mathbf{n}(\mathbf{b}_r + 0.1h_R(\mathbf{b}_r))$. These adjustments follow training on the native $\mathbf{b}_e, \mathbf{b}_r$. For either endpoint x , the complete representation and score are

$$\mathbf{z}_x = [\sqrt{0.6}\mathbf{u}_x; \sqrt{0.4}\mathbf{a}_x], \quad S(r, e) = \mathbf{z}_r^\top \mathbf{z}_e = 0.6\mathbf{u}_r^\top \mathbf{u}_e + 0.4\mathbf{a}_r^\top \mathbf{a}_e. \quad (7)$$

The $512 + |\mathcal{R}_{\text{tr}}|$ dimensional vectors support both directions without direction-specific heads or reranking. Each target training split has independently fitted retrieval weights and dictionary statistics (Appendix H).

REFERENCES

- Keir Adams, Lagnajit Pattanaik, and Connor W. Coley. Learning 3d representations of molecular chirality with invariance to bond rotations. In *International Conference on Learning Representations*, 2022. URL <https://openreview.net/forum?id=hm2tNDdgaFK>.
- Ahmed Elnaggar, Michael Heinzinger, Christian Dallago, Ghalia Rehawi, Yu Wang, Llion Jones, Tom Gibbs, Tamas Feher, Christoph Angerer, Martin Steinegger, Debsindhu Bhownmik, and Burkhard Rost. ProtTrans: Toward understanding the language of life through self-supervised learning. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 44(10):7112–7127, 2022. doi: 10.1109/TPAMI.2021.3095381. URL <https://arxiv.org/abs/2007.06225>.
- Chengyue Gong, Daiheng Zhang, Jeffrey Ouyang-Zhang, Qiang Liu, Adam Klivans, and Daniel J. Diaz. Enzyme classification via semi-supervised functional residue learning. *bioRxiv*, 2026. doi: 10.64898/2026.02.11.705200. URL <https://doi.org/10.64898/2026.02.11.705200>.
- Xiaohong Ji, Zhen Wang, Zhifeng Gao, Hang Zheng, Linfeng Zhang, Guolin Ke, and Weinan E. Uni-Mol2: Exploring molecular pretraining model at scale. In *Advances in Neural Information Processing Systems*, volume 37, 2024. URL <https://arxiv.org/abs/2406.14969>.
- Tatsuya Sagawa. ReactionT5v2: Official model repository. <https://github.com/sagawatatsuya/ReactionT5v2>, n.d. Accessed September 21, 2026.
- Tatsuya Sagawa and Ryosuke Kojima. ReactionT5: A large-scale pre-trained model towards application of limited reaction data. *arXiv preprint arXiv:2311.06708*, 2023. URL <https://arxiv.org/abs/2311.06708>.

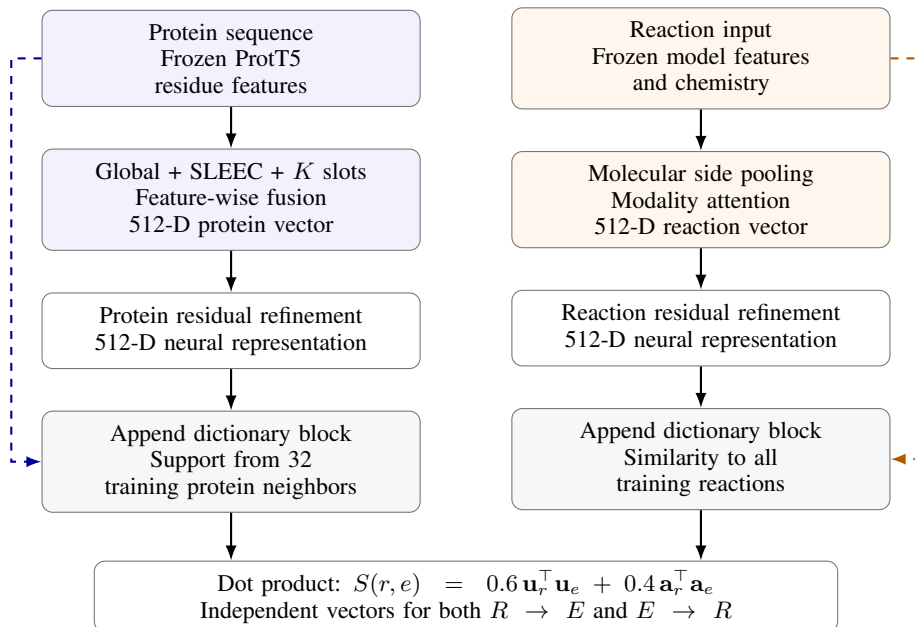


Figure 1: Inference structure of V4, with $K = 4$ by default. Dashed paths supply frozen raw features to the dictionary maps. Dictionary coordinates are computed from frozen raw features and training associations, independently of the neural refinement heads. The arrows into the appended representations indicate composition, not that dictionary features are generated from the heads. The residual heads are trained with association cross-entropy and an identity penalty. The protein fused-view multiplier and refinement multiplier are specified in Section A.8.

A DETAILED ARCHITECTURE AND TRAINING OBJECTIVES

A.1 NOTATION AND COMPUTATIONAL STRUCTURE

We provide the complete definitions underlying Section 3. We use the same symbols as the main text: bold letters denote vectors or matrices, and plain indexed letters their scalar entries. Let $\mathcal{R}_{\text{tr}} = \{r_1, \dots, r_{N_R}\}$ and $\mathcal{E}_{\text{tr}} = \{e_1, \dots, e_{N_E}\}$ denote the unique training endpoints, with association graph $\mathcal{D}_{\text{tr}} \subseteq \mathcal{R}_{\text{tr}} \times \mathcal{E}_{\text{tr}}$ and binary adjacency $Y_{ij}^{\text{tr}} = \mathbb{I}[(r_i, e_j) \in \mathcal{D}_{\text{tr}}]$. Figure 1 summarizes the independently computable representations. Their score measures retrieval compatibility, rather than a calibrated probability of catalytic activity. Both training stages use enzyme–reaction associations; added annotation losses are evaluated only in the ablation in Section A.6.

Throughout, $\text{n}(\mathbf{x}) = \mathbf{x} / \max(\|\mathbf{x}\|_2, \epsilon)$ denotes numerical ℓ_2 normalization, $[x]_+ = \max(0, x)$, $[\mathbf{x}; \mathbf{y}]$ denotes concatenation, and LN denotes layer normalization. All attention distributions exclude padding. Networks with different labels have independent parameters unless sharing is explicitly stated.

A.2 PROTEIN ENCODER: GLOBAL, SLEEC, AND LEARNED RESIDUE VIEWS

For an enzyme sequence e with L retained residues, a frozen ProtT5 model provides contextual residue vectors $\mathbf{H}_e = [\mathbf{h}_1, \dots, \mathbf{h}_L]^T \in \mathbb{R}^{L \times 1024}$ (Elnaggar et al., 2022). Here $\mathcal{V}_e = \{1, \dots, L\}$ indexes valid residues; e is suppressed on per-residue vectors and attention weights. We construct $K + 2$ protein views: one global view, one view defined by a frozen SLEEC functional-residue scorer (Gong et al., 2026), and K learned residue views, with $K = 4$ by default. The scorer is a pretrained resource; it is not fitted to each retrieval target in phase 1.

Global and SLEEC views. The global view is the mean of the retained residue embeddings,

$$\mathbf{x}_e^P = \frac{1}{L} \sum_{i=1}^L \mathbf{h}_i. \quad (8)$$

The frozen scorer computes a scalar logit ℓ_i for each residue using a $1024 \rightarrow 256 \rightarrow 1$ MLP with a ReLU hidden activation. We center these logits within the protein, clip them, and form a smoothed attention distribution:

$$\bar{\ell} = L^{-1} \sum_{i=1}^L \ell_i, \quad \tilde{\ell}_i = \text{clip}(\ell_i - \bar{\ell}, -10, 10), \quad (9)$$

$$p_i^S = \frac{\exp(\tilde{\ell}_i)}{\sum_{j=1}^L \exp(\tilde{\ell}_j)}, \quad a_i^S = (1 - \eta)p_i^S + \frac{\eta}{L}, \quad \eta = 0.05, \quad (10)$$

$$\mathbf{x}_e^S = \sum_{i=1}^L a_i^S \mathbf{h}_i. \quad (11)$$

The vector $\mathbf{x}_e^S \in \mathbb{R}^{1024}$ is a protein-level summary: it averages the residue embeddings $\mathbf{h}_i$ with nonnegative weights a_i^S that sum to one. The superscripts P and S identify the global ProtT5 and SLEEC-weighted features, respectively; superscripts $k \in \{1, \dots, K\}$ identify the learned features. The threshold used for SLEEC diagnostics does not select or remove residues in this encoder.

Learned residue views. We augment the global and SLEEC summaries with K attention-pooled views learned from enzyme–reaction associations. These views provide adaptive pooling of contextual residue features before feature-wise fusion.

A shared residue adapter maps each contextual residue to $d_h = 256$ dimensions:

$$\tilde{\mathbf{h}}_i = \text{GELU}(\mathbf{W}_{\text{res}} \text{LN}(\mathbf{h}_i) + \mathbf{b}_{\text{res}}), \quad \mathbf{k}_i = \text{n}(\mathbf{W}_{\text{key}} \tilde{\mathbf{h}}_i), \quad \mathbf{v}_i = \mathbf{W}_{\text{val}} \tilde{\mathbf{h}}_i + \mathbf{b}_{\text{val}}. \quad (12)$$

The unit-normalized key $\mathbf{k}_i$ determines residue i ’s similarity to a query; the value $\mathbf{v}_i$ contains the features aggregated into the view. The key projection $\mathbf{W}_{\text{key}}$ is bias-free, while the value projection is affine. Queries $\mathbf{q}_k \in \mathbb{R}^{256}$ are orthogonally initialized model parameters shared across enzymes and optimized during phase 1. Normalizing each query gives $\hat{\mathbf{q}}_k = \text{n}(\mathbf{q}_k)$. Query k produces raw view $\mathbf{x}_e^k$:

$$\gamma = 10 \text{sigmoid}(\rho_\gamma), \quad \gamma_{\text{init}} = 4, \quad (13)$$

$$p_{ki} = \frac{\exp(\gamma \hat{\mathbf{q}}_k^\top \mathbf{k}_i)}{\sum_{j=1}^L \exp(\gamma \hat{\mathbf{q}}_k^\top \mathbf{k}_j)}, \quad a_{ki} = (1 - \eta)p_{ki} + \frac{\eta}{L}, \quad (14)$$

$$\mathbf{x}_e^k = \sum_{i=1}^L a_{ki} \mathbf{v}_i, \quad k \in \{1, \dots, K\}. \quad (15)$$

For each query, the softmax is taken over valid residues, so $p_{ki} \geq 0$ and $\sum_i p_{ki} = 1$. The learned scalar ρ_γ parameterizes a bounded scale $\gamma \in (0, 10)$: larger values concentrate attention on residues with higher query–key cosine similarity. The uniform mixture gives $a_{ki} \geq \eta/L$ and $\sum_i a_{ki} = 1$. With $\eta = 0.05$, the resulting view is a mixture of the attention-weighted mean (95%) and the uniform mean (5%) of the same value vectors. These weights quantify feature aggregation, not probabilities that residues are catalytic.

Thus, $\mathbf{q}_k$ defines a shared pooling function, while a_{ki} and $\mathbf{x}_e^k$ are enzyme-specific. The phase-1 objective updates the queries, adapter, and projections through the fused enzyme embedding; entropy and diversity penalties discourage overly concentrated or redundant attention (Section A.4). Views can pool distant residues and overlap, without assigned biochemical roles. Because pooling and fusion use only enzyme features, the views and their fused representation can be cached once the encoder is frozen and reused across candidate reactions. SLEEC contributes a separate view, not a bias to the learned attention logits.

Choice and interpretation of the view count. The view count K sets the number of adaptive summaries available before fusion. With $K = 1$, all learned residue information passes through one summary; larger K can retain complementary patterns, but also increases query, projection, and gate parameters. Views may overlap, so a larger count does not guarantee additional useful information. We retain the original $K = 4$ default as a compromise across screening and bidirectional retrieval: the completed $K \in \{1, 2, 4, 8\}$ ablation finds metric-dependent trade-offs rather than a universally best count (Appendix J). This single-seed study does not establish an optimal capacity or imply that four views correspond to four functional regions or biochemical properties. The residue queries remain frozen throughout phase-2 refinement.

Feature-wise gated fusion. The extracted enzyme features are combined by feature-wise gated fusion. Each feature is independently projected into a common 256-dimensional space using a network of the form

$$F_{a \rightarrow b}(\mathbf{x}) = \mathbf{W}_2 \text{Drop}_{0.1}(\text{GELU}(\mathbf{W}_1 \text{LN}(\mathbf{x}) + \mathbf{b}_1)) + \mathbf{b}_2, \quad (16)$$

whose hidden width is 512 and whose training dropout probability is 0.1. The input width is $a = 1024$ for views P, S and $a = 256$ for learned views. Define $\mathbf{y}_e^v = \mathbf{n}(F_v(\mathbf{x}_e^v))$ for $v \in \mathcal{I} = \{P, S, 1, \dots, K\}$, and let $V = |\mathcal{I}| = K + 2$. A $256V \rightarrow 256 \rightarrow 256V$ gate MLP with GELU maps their concatenation to logits $\mathbf{G}_e \in \mathbb{R}^{V \times 256}$, with rows ordered as $P, S, 1, \dots, K$. The gate softmax is taken over views *separately for each feature coordinate* $d = 1, \dots, 256$:

$$\omega_{e,v,d} = (1 - \zeta) \frac{\exp(G_{e,v,d})}{\sum_{u \in \mathcal{I}} \exp(G_{e,u,d})} + \frac{\zeta}{K + 2}, \quad \zeta = 0.05, \quad (17)$$

$$\bar{\mathbf{y}}_{e,d} = \sum_{v \in \mathcal{I}} \omega_{e,v,d} \mathbf{y}_{e,d}^v. \quad (18)$$

For each enzyme and coordinate, $\sum_v \omega_{e,v,d} = 1$: each fused coordinate is a weighted average of that coordinate across the projected views. The weights vary with both the enzyme and the coordinate, allowing different views to supply different features. The final gate layer is initialized to zero, giving equal initial view weights. The floor is $\zeta/(K + 2)$ per view and coordinate, hence $\zeta/6$ at the default $K = 4$.

Two additional networks of the form in Eq. (16) produce separately normalized global and fused outputs in $\mathbb{R}^{512}$:

$$\mathbf{g}_e = \mathbf{n}(F_g(\mathbf{x}_e^P)), \quad \mathbf{f}_e = \mathbf{n}(F_f(\bar{\mathbf{y}}_e)), \quad (19)$$

$$s = 0.5 \text{sigmoid}(\rho_s), \quad \mathbf{b}_e = \mathbf{n}(\mathbf{g}_e + s\mathbf{f}_e). \quad (20)$$

Here $\mathbf{g}_e$ is computed directly from the sequence mean, while $\mathbf{f}_e$ is computed from the combined views. Their separate projections both output 512-dimensional vectors, making the final addition well-defined. The scalar s is shared across proteins, is initialized to 0.1, and remains strictly between 0 and 0.5. It controls the strength of the entire fused branch; the weights $\omega_{e,v,d}$ instead control the contributions of individual views within that branch. The direct global branch retains sequence context alongside the information supplied by fusion.

Sequence order and interpretation. No additional positional encoding is applied in the pooling module. Its queries attend to the contextual content of ProtT5 vectors. Permuting already computed residue vectors and their masks together leaves this pooling operation unchanged, whereas changing the amino-acid sequence can change the contextual vectors. Neither a slot nor a coordinate of the final 512-dimensional vector is assigned a named catalytic function. Attention weights are learned aggregation weights, not validated active-site labels.

A.3 REACTION ENCODER: MOLECULAR SIDES AND MODALITY FUSION

We use four feature sources: a 768-dimensional ReactionT5v2 reaction embedding (Sagawa & Kojima, 2023; Sagawa, n.d.), 768-dimensional Uni-Mol2 molecule embeddings (Ji et al., 2024), 256-dimensional ChIRo molecule embeddings (Adams et al., 2022), and a 617-dimensional fixed chemistry vector defined in Appendix D. Its four blocks summarize participant substructures, molecular properties, set statistics, and cofactor presence. Their feature extractors are frozen; molecular pooling, modality projections, and fusion are trainable.

Side-specific molecule pooling. For one reaction (with index r suppressed), molecular modality $m \in \{\text{UniMol2}, \text{ChIRo}\}$, and side $s \in \{-, +\}$, let $\mathbf{x}_{s,j}^m$ be the embedding of molecule j . Independent linear attention scorers for the two sides produce

$$\pi_{s,j}^m = \frac{\exp((\mathbf{w}_s^m)^\top \mathbf{x}_{s,j}^m + b_s^m)}{\sum_t \exp((\mathbf{w}_s^m)^\top \mathbf{x}_{s,t}^m + b_s^m)}, \quad \mathbf{p}_s^m = \sum_j \pi_{s,j}^m \mathbf{x}_{s,j}^m. \quad (21)$$

The configured `directional_delta` composition is

$$\mathbf{c}_r^m = [\mathbf{p}_-^m; \mathbf{p}_+^m; \mathbf{p}_+^m - \mathbf{p}_-^m; |\mathbf{p}_+^m - \mathbf{p}_-^m|]. \quad (22)$$

It has dimension 3072 for Uni-Mol2 and 1024 for ChIRo. Direction is represented by the ordered sides and signed difference when physical sides are supplied. The first two blocks preserve the pooled molecular context of each side; the last two expose the sign and magnitude of coordinate-wise contrasts. These contrasts are between learned summaries, rather than an explicit atom-mapped list of bond changes. Only Uni-Mol2 and ChIRo use this pooling: ReactionT5v2 and the fixed chemistry descriptor already provide one vector per reaction.

Input semantics. For EnzymeMap, inputs retain their physical reactant and product sides. ReactZyme’s released participant-set inputs use the established self-reaction adapter, placing the same complete set on both sides. The two side poolers still have different parameters, so their pooled vectors need not be equal. Their difference on these self-reactions must not be interpreted as an observed chemical transformation direction. The choice of chemical input representation is separate from the choice of retrieval direction.

Modality tokens and attention. For $m \in \{\text{T5}, \text{UniMol2}, \text{ChIRo}, \text{chem}\}$, let $\mathbf{c}_r^m$ denote the corresponding reaction input. Each independent modality MLP has widths $d_m \rightarrow 1024 \rightarrow 512 \rightarrow 512$, with ReLU after the first two linear layers. A shared layer-normalization module then gives the modality token

$$\mathbf{t}_r^m = \text{LN}(\text{MLP}_m(\mathbf{c}_r^m)) \in \mathbb{R}^{512}. \quad (23)$$

An attention MLP of widths $512 \rightarrow 512 \rightarrow 1$, with a ReLU hidden activation, scores the tokens. If $\mathcal{M}_r$ is the set of available modalities,

$$\alpha_r^m = \frac{\exp(A(\mathbf{t}_r^m))}{\sum_{n \in \mathcal{M}_r} \exp(A(\mathbf{t}_r^n))}, \quad \mathbf{t}_r = \sum_{m \in \mathcal{M}_r} \alpha_r^m \mathbf{t}_r^m, \quad \mathbf{b}_r = \mathbf{n}(F_{\text{out}}(\mathbf{t}_r)). \quad (24)$$

The output MLP F_{out} has widths $512 \rightarrow 1024 \rightarrow 512$ and a ReLU hidden activation. Unavailable modalities are masked out of attention. Molecule attention therefore combines molecules within one modality and side, whereas modality attention combines the four reaction-level tokens. Reaction modality dropout and projection dropout are zero in the reported recipe. The resulting $\mathbf{b}_r$ and $\mathbf{b}_e$ inhabit a shared 512-dimensional retrieval space.

A.4 PHASE 1: ANCHOR-BALANCED, DECOUPLED MULTI-POSITIVE ALIGNMENT

Training minibatches sample association rows and deduplicate their endpoint identifiers before constructing a reaction-by-enzyme score matrix. Let $\mathcal{R}_B$ and $\mathcal{E}_B$ be these distinct endpoints. Positives are identified from *all known training associations between endpoints in the batch*, rather than only the association rows sampled for that batch. For a reaction anchor r , define

$$P_B(r) = \{e \in \mathcal{E}_B : (r, e) \in \mathcal{D}_{\text{tr}}\}, \quad N_B(r) = \mathcal{E}_B \setminus P_B(r), \quad (25)$$

and define $P_B(e)$ and $N_B(e)$ analogously for enzyme anchors. Only anchors with at least one positive and one remaining candidate enter the respective directional average.

The logits are $s_{re}^{(1)} = \mathbf{b}_r^\top \mathbf{b}_e / \tau_1$ with $\tau_1 = 0.2$. For each known positive p of an anchor a , the decoupled loss is

$$\ell_{\text{dec}}(a, p) = -\log \frac{\exp(s_{ap}^{(1)})}{\exp(s_{ap}^{(1)}) + \sum_{n \in N_B(a)} \exp(s_{an}^{(1)})} \quad (26)$$

$$= \log \left(1 + \sum_{n \in N_B(a)} \exp(s_{an}^{(1)} - s_{ap}^{(1)}) \right). \quad (27)$$

Other known positives are excluded from the denominator for this individual positive. Thus every positive must compete with the remaining candidates, without competing with its known positive partners. This differs from a single log-sum over all positives, which can be dominated by one easy positive.

Let $\mathcal{A}_R$ and $\mathcal{A}_E$ be the valid reaction and enzyme anchors in the batch. We first average over positives within an anchor, then average over anchors, and finally average the two directions:

$$\mathcal{L}_{R \rightarrow E}^{(1)} = \frac{1}{|\mathcal{A}_R|} \sum_{r \in \mathcal{A}_R} \frac{1}{|P_B(r)|} \sum_{e \in P_B(r)} \ell_{\text{dec}}(r, e), \quad (28)$$

$$\mathcal{L}_{E \rightarrow R}^{(1)} = \frac{1}{|\mathcal{A}_E|} \sum_{e \in \mathcal{A}_E} \frac{1}{|P_B(e)|} \sum_{r \in P_B(e)} \ell_{\text{dec}}(e, r), \quad (29)$$

$$\mathcal{L}_{\text{align}}^{(1)} = \frac{1}{2} \mathcal{L}_{R \rightarrow E}^{(1)} + \frac{1}{2} \mathcal{L}_{E \rightarrow R}^{(1)}. \quad (30)$$

A direction with no valid anchor contributes zero. The normalization prevents an anchor’s contribution from growing simply in proportion to its number of positives in the matrix. It does not make anchors equally frequent over an epoch when association rows are sampled. Unobserved pairs are still used as contrastive negatives with weight 1; they are not measured inactive pairs.

Attention regularization. The protein encoder also returns two small penalties on the *raw* learned-slot distributions p_{ki} , before the uniform mixture in Eq. (15). For a protein of length L , define

$$H_k = - \sum_i p_{ki} \log p_{ki}, \quad H_{\min}(L) = \min\{\frac{1}{2} \log L, \log 4\}, \quad (31)$$

$$E(e) = \frac{1}{K} \sum_{k=1}^K [H_{\min}(L) - H_k]_+^2. \quad (32)$$

This penalty discourages attention with very small effective support. To discourage nearly identical attention patterns, center each raw distribution against uniform attention: $\bar{\mathbf{p}}_k = \mathbf{p}_k - L^{-1} \mathbf{1}$, where $\mathbf{p}_k = (p_{ki})_{i \in \mathcal{V}_e}$. For eligible unordered pairs $\mathcal{Q}_e = \{(k, l) : k < l, \|\bar{\mathbf{p}}_k\|_2 > 10^{-6}, \|\bar{\mathbf{p}}_l\|_2 > 10^{-6}, L > 1\}$, let

$$D(e) = \frac{1}{\max(1, |\mathcal{Q}_e|)} \sum_{(k, l) \in \mathcal{Q}_e} \left[\frac{\bar{\mathbf{p}}_k^\top \bar{\mathbf{p}}_l}{\|\bar{\mathbf{p}}_k\|_2 \|\bar{\mathbf{p}}_l\|_2} - 0.9 \right]_+^2. \quad (33)$$

The main-text penalties are $\mathcal{L}_{\text{ent}} = |\mathcal{E}_B|^{-1} \sum_{e \in \mathcal{E}_B} E(e)$ and $\mathcal{L}_{\text{div}} = |\mathcal{E}_B|^{-1} \sum_{e \in \mathcal{E}_B} D(e)$. Uniform heads have no meaningful centered direction and are excluded from this comparison. The complete phase-1 objective is

$$\mathcal{L}^{(1)} = \mathcal{L}_{\text{align}}^{(1)} + 0.01 \frac{1}{|\mathcal{E}_B|} \sum_{e \in \mathcal{E}_B} E(e) + 0.01 \frac{1}{|\mathcal{E}_B|} \sum_{e \in \mathcal{E}_B} D(e). \quad (34)$$

The safeguards in Eqs. (32) and (33) encourage numerical and representational diversity; they do not establish a biological interpretation of the attention heads.

A.5 PHASE 2: FULL-GRAPH RESIDUAL REFINEMENT

After retaining the epoch-10 phase-1 snapshot, we freeze both F3 encoders and cache their unadjusted outputs $\mathbf{b}_r, \mathbf{b}_e$ for the unique training endpoints. Two independent residual heads h_R, h_E each comprise layer normalization, a $512 \rightarrow 1024$ linear layer, GELU, and a $1024 \rightarrow 512$ linear layer. The last layer’s weights and biases are initialized to zero. During phase-2 training,

$$\mathbf{u}_r^{\text{tr}} = \text{n}(\mathbf{b}_r + \kappa h_R(\mathbf{b}_r)), \quad \mathbf{u}_e^{\text{tr}} = \text{n}(\mathbf{b}_e + \kappa h_E(\mathbf{b}_e)), \quad \kappa = 0.2. \quad (35)$$

Consequently, both heads initially preserve the normalized base representations. No gradient is propagated to the frozen F3 encoders.

Uniform-positive cross-entropy over the complete graph. We abbreviate $s_{ij}^{(2)} = s_{r_i e_j}^{(2)} = (\mathbf{u}_{r_i}^{\text{tr}})^\top \mathbf{u}_{e_j}^{\text{tr}} / \tau_2$, with $\tau_2 = 0.2$. Let $d_i^R = \sum_j Y_{ij}^{\text{tr}}$ and $d_j^E = \sum_i Y_{ij}^{\text{tr}}$. Every retained training endpoint has at least one association. The directional objectives are

$$\mathcal{L}_{R \rightarrow E}^{(2)} = \frac{1}{N_R} \sum_i \left[\log \sum_{j=1}^{N_E} \exp(s_{ij}^{(2)}) - \frac{1}{d_i^R} \sum_j Y_{ij}^{\text{tr}} s_{ij}^{(2)} \right], \quad (36)$$

$$\mathcal{L}_{E \rightarrow R}^{(2)} = \frac{1}{N_E} \sum_j \left[\log \sum_{i=1}^{N_R} \exp(s_{ij}^{(2)}) - \frac{1}{d_j^E} \sum_i Y_{ij}^{\text{tr}} s_{ij}^{(2)} \right], \quad (37)$$

$$\mathcal{L}_{\text{CE}}^{(2)} = \frac{1}{2} \mathcal{L}_{R \rightarrow E}^{(2)} + \frac{1}{2} \mathcal{L}_{E \rightarrow R}^{(2)}. \quad (38)$$

These are cross-entropies with a uniform target distribution over each anchor’s known positives. Unlike phase 1, the denominator includes all opposite-type training endpoints, including all positives. Phase 2 therefore changes both the coverage of the contrastive bank and the form of the loss; it is not simply Eq. (27) evaluated on a larger batch.

Preserving the original geometry. An identity penalty limits changes to the frozen base representations:

$$\mathcal{L}_{\text{id}} = \frac{1}{2N_R} \sum_i [1 - (\mathbf{u}_{r_i}^{\text{tr}})^\top \mathbf{n}(\mathbf{b}_{r_i})] + \frac{1}{2N_E} \sum_j [1 - (\mathbf{u}_{e_j}^{\text{tr}})^\top \mathbf{n}(\mathbf{b}_{e_j})]. \quad (39)$$

The complete refinement objective is

$$\boxed{\mathcal{L}^{(2)} = \mathcal{L}_{\text{CE}}^{(2)} + 10\mathcal{L}_{\text{id}}.} \quad (40)$$

Both heads are optimized jointly for 100 updates, each of which includes the entire compact training graph. Added biological-label losses are not part of this objective.

A.6 BIOLOGICAL-SUPERVISION ABLATION

This ablation is excluded from the proposed method. We retain its definition to document the mixed experimental outcomes. The ablated variant, V4+Bio, adds supervision to the same refinement heads. It constrains relationships among reaction representations and, separately, among enzyme representations. The association loss connects the two endpoint types. No classification heads, biological feature coordinates, additional residue queries, or inference-time annotation inputs are introduced.

Annotation families and confidence. We use three families $\mathcal{F} = \{\text{EC}, \text{cof}, \text{mech}\}$. Each training endpoint can belong to multiple categories. A membership in category c has confidence $w_{ic} \in (0, 1]$; missing memberships create no annotation evidence. For EC, each specified prefix is retained. A category at hierarchy depth $d \in \{1, 2, 3, 4\}$ has weight $\xi_c \in \{0.125, 0.25, 0.5, 1\}$, respectively. Unknown EC digits are not imputed. Cofactor-associated categories are recognized from reaction participants and assigned confidence 0.4. They describe participant presence, which is an imperfect proxy for a biochemical cofactor requirement. Mechanism categories are coarse descriptors derived from atom-mapped bond changes; mappings below confidence 0.5 are omitted. These descriptors do not specify complete catalytic mechanisms. Cofactor and mechanism category weights are $\xi_c = 1$.

Reaction annotations are transferred to proteins only along training associations. When several observations support the same membership, the maximum confidence is retained. Native protein EC annotations can additionally describe multiple functions of a training protein. Appendix G specifies the different annotation sources for the two benchmarks.

Relative within-category and background distances. Consider one endpoint type $X \in \{R, E\}$ and one family f . Let $\hat{\mathbf{u}}_i = \mathbf{n}(\mathbf{u}_i^{\text{tr}})$ and $\delta_{ij} = 1 - \hat{\mathbf{u}}_i^\top \hat{\mathbf{u}}_j$. The index i refers to a training endpoint of type X ; these 512-dimensional vectors differ from the complete retrieval vectors $\mathbf{z}_x$. Let $\mathcal{C}_{X,f}$ contain the categories of family f for endpoint type X . Let I_c index endpoints annotated with category c , $n_c = |I_c|$, and $\mathcal{A}_f = \bigcup_{c \in \mathcal{C}_{X,f}} I_c$. The comparison set is $B_c = \mathcal{A}_f \setminus I_c$. An endpoint

without any annotation in f is excluded from both membership and background for that family. For a multiply annotated background endpoint, set $\bar{w}_j^f = \max_{c' \in \mathcal{C}_{X,f}: j \in I_{c'}} w_{jc'}$.

For eligible categories, define the within-category and background confidence masses

$$T_c = \sum_{\substack{i,j \in I_c \\ i \neq j}} w_{ic} w_{jc}, \quad U_c = \sum_{i \in I_c} \sum_{j \in B_c} w_{ic} \bar{w}_j^f, \quad (41)$$

and confidence-weighted mean distances

$$d_c^{\text{within}} = \frac{1}{T_c} \sum_{\substack{i,j \in I_c \\ i \neq j}} w_{ic} w_{jc} \delta_{ij}, \quad (42)$$

$$d_c^{\text{background}} = \frac{1}{U_c} \sum_{i \in I_c} \sum_{j \in B_c} w_{ic} \bar{w}_j^f \delta_{ij}. \quad (43)$$

The per-category penalty is

$$\ell_c = \bar{w}_c [m + d_c^{\text{within}} - d_c^{\text{background}}]_+, \quad m = 0.1. \quad (44)$$

Here $\bar{w}_c = T_c / [n_c(n_c - 1)]$ is the mean pairwise annotation confidence. Its prefactor preserves the strength of annotation confidence. For example, if every membership in a category has confidence 0.4, this prefactor is $0.4^2 = 0.16$. Computing only confidence-normalized distances would remove that common confidence reduction.

A category contributes only if $n_c \geq 2$, $|B_c| \geq 1$, $T_c > 0$, and $U_c > 0$. Let $\mathcal{C}_{X,f}^*$ be the eligible categories. We use a category-weighted average, followed by a fixed average over endpoint types:

$$\mathcal{L}_{X,f} = \frac{\sum_{c \in \mathcal{C}_{X,f}^*} \xi_c \ell_c}{\sum_{c \in \mathcal{C}_{X,f}^*} \xi_c}, \quad \mathcal{L}_f = \frac{1}{2} (\mathcal{L}_{R,f} + \mathcal{L}_{E,f}). \quad (45)$$

An empty eligible set contributes zero. Averaging by category, rather than by the total number of pairs across categories, limits domination by very large annotation groups. Membership overlap is allowed; the background for one category can share other categories with its members.

Objective of the biological ablation. The ablation augments Eq. (40) as

$$\mathcal{L}_{\text{Bio}}^{(2)} = \mathcal{L}^{(2)} + \frac{1}{3} \sum_{f \in \mathcal{F}} \mathcal{L}_f. \quad (46)$$

All three families have unit coefficients before the fixed average. This recipe is recorded as `all_1` in the implementation. In family-removal ablation controls, the corresponding term is omitted; the divisor remains three. Shuffled-label controls preserve annotation profiles, confidences, and category frequencies while permuting their assignment to training endpoints.

The relative loss expresses a geometric preference for annotated peers. The background does not assert that a protein is inactive for a reaction, and different categories are not mutually exclusive biological functions. Incomplete annotations and enzyme promiscuity can make these geometric references imperfect. The relative hinge does not by itself guarantee non-collapsed embeddings; association supervision remains necessary. In the second training stage, biological gradients reach the residual heads but not the frozen residue queries or feature extractors. Exact category-sum identities avoid constructing a dense biological pairwise matrix (Appendix E).

A.7 TRAINING-REACTION DICTIONARY REPRESENTATION

In addition to the neural embedding, both endpoints receive a vector $\mathbf{a} \in \mathbb{R}^{N_R}$ whose coordinates index the target’s training reactions. The dictionary uses frozen raw features and observed training associations. It is fitted once for each target split and is independent of the current query or candidate collection.

Training-only centering. For a raw feature family with available training values $\mathbf{x}_i$, we compute

$$\boldsymbol{\mu} = \frac{1}{N_{\text{avail}}} \sum_{i \in \text{available train}} n(\mathbf{x}_i), \quad \psi(\mathbf{x}) = n(n(\mathbf{x}) - \boldsymbol{\mu}). \quad (47)$$

The order is normalization, centering, then normalization. Enzyme features are cached ProtT5 residue means. Reaction features concatenate four separately centered and normalized blocks (ReactionT5v2, Uni-Mol2, ChIRo, chemistry), followed by normalization of the concatenation. Unavailable blocks are zero. For the dictionary’s molecular blocks, the implemented export uses the *mean forward-reactant bank*: this is the complete participant set for ReactZyme self-reactions and the reactant side for EnzymeMap. These fixed raw features differ from the trainable side pooling in the neural reaction tower.

Enzyme-to-training-reaction transfer. Let $\psi_E(e)$ be the centered enzyme feature and let $\mathcal{N}_{32}(e)$ be its 32 nearest training proteins by cosine similarity $c_{ej} = \psi_E(e)^\top \psi_E(e_j)$. With $\tau_E = 0.03$, define

$$w_{ej}^{\text{NN}} = \exp\left(\frac{c_{ej} - \max_{t \in \mathcal{N}_{32}(e)} c_{et}}{\tau_E}\right), \quad (48)$$

$$\tilde{a}_{e,i} = \max_{\substack{j \in \mathcal{N}_{32}(e) \\ Y_{ij}^{\text{tr}}=1}} w_{ej}^{\text{NN}}, \quad \mathbf{a}_e = n(\tilde{\mathbf{a}}_e). \quad (49)$$

An empty maximum is zero. The maximum merges repeated support for a training reaction without summing the frequency of similar supporting proteins. There is no exclusion of an exact training protein when it is an eligible neighbor. Such information is permitted by the benchmark’s training graph.

Reaction-to-training-reaction response. Let $\psi_R(r)$ denote the normalized raw reaction feature and $c_{ri} = \psi_R(r)^\top \psi_R(r_i)$. We use all training reaction anchors:

$$\tilde{a}_{r,i} = \exp\left(\frac{c_{ri} - \max_t c_{rt}}{\tau_R}\right), \quad \mathbf{a}_r = n(\tilde{\mathbf{a}}_r), \quad \tau_R = 0.03. \quad (50)$$

The implementation floors full-support exponential responses at the smallest positive normal FP32 value before normalization. The rowwise maximum in Eqs. (49)–(50) stabilizes exponentiation and otherwise cancels under normalization. Full reaction-anchor support avoids exact support mismatch caused by truncating both endpoint vectors to separate nearest-anchor sets. It does not ensure informative transfer for a reaction remote from the training chemistry.

A.8 INFERENCE ADJUSTMENTS AND FINAL BIDIRECTIONAL SCORE

The reported recipe makes two explicit inference adjustments after both training stages. First, it strengthens the entire fused protein branch:

$$\tilde{\mathbf{b}}_e = n(\mathbf{g}_e + \nu s \mathbf{f}_e), \quad \nu = 3, \quad \tilde{\mathbf{b}}_r = \mathbf{b}_r. \quad (51)$$

The scalar νs is checked to remain below 0.5 for these checkpoints. This multiplies the complete fused-view contribution, including its global, SLEEC, and learned-slot information. It does not multiply only SLEEC weights. Phase 2 was fitted to the native $\mathbf{b}_e$, before this adjustment.

Second, the existing residual heads are applied with inference multiplier $c = 0.5$:

$$\mathbf{u}_e = n(\tilde{\mathbf{b}}_e + c\kappa h_E(\tilde{\mathbf{b}}_e)), \quad \mathbf{u}_r = n(\mathbf{b}_r + c\kappa h_R(\mathbf{b}_r)), \quad c\kappa = 0.1. \quad (52)$$

The multiplier halves the residual coefficient relative to training. It is not a hard bound on the update norm or on the resulting angular displacement. The dictionary representation is unchanged by either adjustment.

With semantic weight $\alpha = 0.4$, define independent endpoint embeddings

$$\mathbf{z}_e = [\sqrt{1-\alpha} \mathbf{u}_e; \sqrt{\alpha} \mathbf{a}_e], \quad \mathbf{z}_r = [\sqrt{1-\alpha} \mathbf{u}_r; \sqrt{\alpha} \mathbf{a}_r]. \quad (53)$$

Their dot product gives the final score

$$S(r, e) = \mathbf{z}_r^\top \mathbf{z}_e = 0.6 \mathbf{u}_r^\top \mathbf{u}_e + 0.4 \mathbf{a}_r^\top \mathbf{a}_e. \quad (54)$$

Thus the neural block has 512 dimensions, whereas the complete representation has $512 + N_R$ dimensions. The same score is used in both retrieval directions. Dense neural blocks and sparse enzyme dictionary blocks can be stored separately without changing the mathematical score. The method requires the frozen feature extractors, trained encoders, refinement heads, training dictionary, and feature-normalization statistics at deployment.

A.9 OPTIMIZATION AND TARGET-SPECIFIC FITTING

We fit a separate model for each ReactZyme split and for EnzymeMap, using the corresponding target training associations. Trainable retrieval components are freshly initialized for each target; the pretrained feature extractors and SLEEC prior are reused and frozen. In particular, EnzymeMap does not inherit a ReactZyme-trained F3 checkpoint.

Phase 1 uses seed 42, association batch size 512, AdamW learning rate 10^{-4} , weight decay 10^{-2} , and constant learning rate. We use the epoch-10 snapshot. The original ReactZyme trajectories continued to epoch 20; their retained epoch-10 snapshots define this recipe. Features are cached, training uses mixed BF16 precision, and contrastive calculations are kept in FP32. Phase 2 uses seed 42, 100 full-graph AdamW updates, learning rate 10^{-4} , weight decay 10^{-3} , and gradient-norm clipping at 1. The declared step-100 state is used for the reported configuration.

The graph, dictionary centers, and dictionary associations use training endpoints only. Added biological annotations are restricted to the separate ablation. Matching target training associations does not imply identical pretrained resources across methods. The present recipe was developed through exploratory experiments; the fixed recipe should not be described as a configuration selected without prior inspection of benchmark outcomes.

B NOTATION AND EXACT LAYER SPECIFICATIONS

Symbol	Meaning
$\mathbf{Y}^{\text{tr}}, \mathcal{D}_{\text{tr}}$	Binary target-training association matrix and its unique positive edges.
$\mathbf{b}_e, \mathbf{b}_r$	Native 512-dimensional phase-1 protein and reaction outputs.
$\mathbf{g}_e, \mathbf{f}_e$	Separately normalized global and fused protein branches.
$K, V = K + 2$	Learned residue-view count (default 4) and total protein-view count.
$i \in \mathcal{V}_e, k, v, d$	Residue index; learned query index $1 \leq k \leq K$; view label $v \in \mathcal{I} = \{P, S, 1, \dots, K\}$; feature coordinate $1 \leq d \leq 256$.
$\mathbf{h}_i, \ell_i, \tilde{\ell}_i$	Frozen 1024-dimensional residue feature, SLEEC logit, and centered/clipped logit; enzyme index suppressed.
$\mathbf{q}_k, \hat{\mathbf{q}}_k$	Shared trainable query and its numerically normalized version, both 256-dimensional.
$\mathbf{k}_i, \mathbf{v}_i$	Normalized residue key and unnormalized value, both 256-dimensional; enzyme index suppressed.
$\mathbf{x}_e^v$	Raw views: $v = P$ global ProtT5 and $v = S$ SLEEC (1024 dimensions); $v = k$ learned query k (256 dimensions).
$\mathbf{y}_e^v, \bar{\mathbf{y}}_e$	Projected, normalized views and their coordinate-wise fused vector, all 256-dimensional.
p_{ki}, a_{ki}	Raw and uniform-smoothed residue attention for learned slot k .
$\omega_{e,vd}$	Gate weight for protein view v and hidden coordinate d .
s	Learned bounded protein fusion scale; initial value 0.1.
ν	Protein fusion multiplier used after training; value 3.
h_E, h_R	Independent residual heads fitted with frozen F3 inputs.
κ, c	Residual coefficient in training (0.2) and inference multiplier (0.5).
$\mathbf{u}_e^{\text{tr}}, \mathbf{u}_r^{\text{tr}}$	Refined endpoints used in the phase-2 training objectives.
$\mathbf{u}_e, \mathbf{u}_r$	Refined endpoints after the two inference adjustments.

Symbol	Meaning
$s_{ax}^{(1)}, s_{ax}^{(2)}$	Scalar association logits in phases 1 and 2; distinct from the fusion scale s and final embeddings $\mathbf{z}_x$.
$\hat{\mathbf{u}}_i$	Renormalized phase-2 vector in the biological-supervision ablation only.
ξ_c, w_{ic}	Category weight and membership confidence in the biological ablation.
$\mathbf{a}_e, \mathbf{a}_r$	Normalized training-reaction dictionary coordinates.
$\mathbf{z}_e, \mathbf{z}_r$	Complete $512 + N_R$ endpoint representations.
α	Dictionary contribution to the final dot product; value 0.4.
$\mathcal{M}_r, \mathcal{P}_r$	Available reaction modalities and multiset of valid participant molecules, respectively.

Table 2: Active neural components in the default $K = 4$ configuration. Dimensions are listed in execution order.

Component	Specification
Frozen residue features	ProtT5; 1024 coordinates per retained residue.
Frozen SLEEC scorer	Linear $1024 \rightarrow 256$; ReLU; linear $256 \rightarrow 1$.
Residue adapter	LayerNorm(1024); linear $1024 \rightarrow 256$; GELU.
Keys and values	Keys: bias-free linear $256 \rightarrow 256$ followed by normalization. Values: affine linear $256 \rightarrow 256$.
Learned residue queries	Four trainable 256-dimensional vectors; orthogonal initialization; normalized before scoring.
Global/SLEEC view projection	Two independent networks: LayerNorm(1024); linear $1024 \rightarrow 512$; GELU; dropout 0.1; linear $512 \rightarrow 256$; normalization.
Learned-slot view projection	Four independent networks: LayerNorm(256); linear $256 \rightarrow 512$; GELU; dropout 0.1; linear $512 \rightarrow 256$; normalization.
Protein gate	Linear $1536 \rightarrow 256$; GELU; linear $256 \rightarrow 1536$; reshape to 6×256 ; softmax over views.
Global output branch	LayerNorm(1024); linear $1024 \rightarrow 512$; GELU; dropout 0.1; linear $512 \rightarrow 512$; normalization.
Fused output branch	LayerNorm(256); linear $256 \rightarrow 512$; GELU; dropout 0.1; linear $512 \rightarrow 512$; normalization.
Uni-Mol2 molecule pooling	Separate scalar affine attention for reactants and products, acting on 768-dimensional molecule vectors.
ChIRo molecule pooling	Separate scalar affine attention for reactants and products, acting on 256-dimensional molecule vectors.
Reaction modality MLPs	Input widths 768, 3072, 1024, 617 for T5, Uni-Mol2 composition, ChIRo composition, and chemistry. Each follows $d_m \rightarrow 1024 \rightarrow 512 \rightarrow 512$; ReLU after its first two layers.
Reaction token normalization	One LayerNorm(512), shared over modality tokens. No per-token ℓ_2 normalization in this configuration.
Reaction modality attention	Linear $512 \rightarrow 512$; ReLU; linear $512 \rightarrow 1$; masked softmax over available modalities.
Reaction output projection	Linear $512 \rightarrow 1024$; ReLU; linear $1024 \rightarrow 512$; normalization.
Phase-2 heads	Two independent copies of LayerNorm(512); linear $512 \rightarrow 1024$; GELU; linear $1024 \rightarrow 512$. Last affine layer initialized to zero.

The two phase-2 heads contain 2,102,272 parameter values in total. The four residue queries are shared across proteins; they are not four trainable vectors created separately for each sequence.

C ATTENTION IMPLEMENTATION AND SEQUENCE PREPROCESSING

Padding is removed from softmax denominators and pooling sums, and its contribution is zero. Before the frozen scorer and residue adapter, padded values are replaced by zero so that padded NaN sentinels do not enter learned linear maps. SLEEC logits are detached. Centering its logits alone would not alter softmax, but centering before clipping gives a protein-relative clipping rule. The use of logits, rather than sigmoid scores, is material to the resulting distribution.

The learned attention scale is initialized by $\rho_\gamma = \text{logit}(4/10)$, and the fusion scalar by $\rho_s = \text{logit}(0.1/0.5)$. The normalization floor in the protein multiview module is 10^{-6} . Attention entropy uses natural logarithms and a probability floor of 10^{-12} inside the logarithm. Regularization in Eqs. (32)–(33) is evaluated on raw learned attention, not on the SLEEC distribution and not on the smoothed distributions used to pool values.

For the F3 protein tower, at most 1022 residue vectors are retained. For a longer cached sequence, the deterministic `ends_center` policy keeps $\lfloor 1022/4 \rfloor$ vectors at each end and the remaining budget from the center, preserving their original order. The dictionary’s enzyme means are exported separately by averaging the stored residue bank; they should not be silently replaced with the neural tower’s potentially truncated mean. EnzymeMap’s specified screening library already restricts sequence length to at most 650 amino acids.

The pooling operations scale linearly in sequence length for a fixed number of slots. In particular, learned residue attention forms an $L \times K$ matrix ($L \times 4$ by default), not an $L \times L$ self-attention matrix. This complexity statement concerns the trainable pooling module after frozen features are available; it does not describe the cost of computing ProtT5 features.

D FIXED CHEMISTRY DESCRIPTOR

The fixed chemistry descriptor gives the reaction encoder an explicit summary of participant substructures, physicochemical properties, and recognized cofactors alongside its pretrained reaction and molecular embeddings. These quantities are available directly from participant SMILES; their construction does not require enzyme–reaction association labels. This supplies observable chemical cues whose contribution to retrieval is learned through the modality projection and fusion.

The 617-dimensional chemistry vector consists of four blocks:

$$617 = 512 + 64 + 4 + 37. \quad (55)$$

The first block averages 512-bit radius-2 Morgan fingerprints of valid participant molecules. The second block applies sum, mean, maximum, and standard-deviation aggregation to 16 molecule descriptors, yielding 64 coordinates. Table 3 lists the properties in their exact order within the per-molecule vector $\mathbf{d}(m)$. Each is computed with RDKit from a valid participant molecule. The aggregate block concatenates all 16 sums, then all 16 means, maxima, and population standard deviations; these are summaries across participants, irrespective of reaction side.

Table 3: The 16 molecular properties aggregated in the reaction chemistry descriptor. Each property contributes four aggregate coordinates, giving $16 \times 4 = 64$ in total. Function names identify the RDKit calculations.

Index	Property	Definition or calculation
1	Exact molecular weight	Exact molecular weight from <code>CalcExactMolWt</code> .
2	Heavy-atom count	Number of atoms with atomic number $Z > 1$.
3	Heteroatom count	Number of atoms with $Z \notin \{0, 1, 6\}$; wildcard atoms, hydrogen, and carbon are excluded.
4	Formal charge	Net molecular formal charge from <code>GetFormalCharge</code> .
5	Hydrogen-bond donors	Donor count from <code>NumHDonors</code> .
6	Hydrogen-bond acceptors	Acceptor count from <code>NumHAcceptors</code> .

Index	Property	Definition or calculation (continued)
7	Topological polar surface area	Molecular TPSA from CalcTPSA.
8	$\log P$	RDKit Crippen estimate from MolLogP.
9	Rotatable-bond count	Number of rotatable bonds from NumRotatableBonds.
10	Ring count	Number of rings from RingCount.
11	Aromatic-heavy-atom fraction	Number of aromatic heavy atoms divided by $\max(N_{\text{heavy}}, 1)$, where N_{heavy} is the heavy-atom count.
12	Atom stereocenter count	Number of centers returned by FindMolChiralCenters, including unassigned centers.
13	Wildcard-atom count	Number of atoms with symbol $*$.
14	Nitrogen count	Number of atoms with symbol N.
15	Oxygen count	Number of atoms with symbol O.
16	Phosphorus/sulfur count	Combined number of atoms with symbols P or S.

The four set statistics are participant count, unique-participant fraction, valid-participant fraction, and wildcard-containing fraction among valid participants. The final block is a 37-position capacity for the recognized core-cofactor vocabulary fitted on training reactions; unused positions are zero. Only the 64 aggregated property coordinates are standardized. Their means and standard deviations are fitted on valid training reactions, with a standard deviation below 10^{-6} replaced by one. Fingerprints, set statistics, and cofactor indicators are not standardized in this construction.

More precisely, let $\mathcal{P}_r$ be the multiset of valid participant molecules, with $n_r = |\mathcal{P}_r| > 0$. For a molecule m , let $\phi(m) \in \{0, 1\}^{512}$ denote its radius-2 Morgan fingerprint and $\mathbf{d}(m) \in \mathbb{R}^{16}$ its property vector. Define

$$\bar{\phi}_r = \frac{1}{n_r} \sum_{m \in \mathcal{P}_r} \phi(m), \quad (56)$$

$$\mathbf{d}_r = \left[\sum_{m \in \mathcal{P}_r} \mathbf{d}(m); \text{mean}_{m \in \mathcal{P}_r} \mathbf{d}(m); \max_{m \in \mathcal{P}_r} \mathbf{d}(m); \text{std}_{m \in \mathcal{P}_r} \mathbf{d}(m) \right], \quad (57)$$

$$\mathbf{c}_r^{\text{chem}} = [\bar{\phi}_r; (\mathbf{d}_r - \boldsymbol{\mu}_{\text{tr}}) \oslash \boldsymbol{\sigma}_{\text{tr}}; \mathbf{s}_r^{\text{set}}; \mathbf{q}_r^{\text{cof}}] \in \mathbb{R}^{617}. \quad (58)$$

Here $\boldsymbol{\mu}_{\text{tr}}, \boldsymbol{\sigma}_{\text{tr}} \in \mathbb{R}^{64}$ are the training means and stabilized standard deviations defined above, and $\oslash$ denotes elementwise division. All property aggregations and standardization are coordinate-wise, and std uses the population convention. The vector $\mathbf{s}_r^{\text{set}} \in \mathbb{R}^4$ contains the four set statistics above; $\mathbf{q}_r^{\text{cof}} \in \{0, 1\}^{37}$ records presence of recognized core cofactors in the training vocabulary. An unseen cofactor does not create a new coordinate. Reactions with no valid participant have zero fingerprint and property blocks and an unavailable-chemistry mask; modality attention excludes this input.

“Fixed” therefore distinguishes feature construction from neural learning. The fingerprint and property definitions, training-derived normalization, and cofactor vocabulary remain unchanged while the reaction encoder is trained. A learned chemistry-specific MLP maps $\mathbf{c}_r^{\text{chem}}$ to a 512-dimensional token, which participates in the same modality attention as the other three sources (Section A.3).

Participant canonicalization removes atom-map indices while retaining isomeric SMILES information. When both sides of a stored self-reaction have the same canonical participant multiset, the participant list is used once. Otherwise, the two sides are combined for this descriptor. Thus this chemistry block is a participant-set feature, even when the neural molecular branches retain physical reaction sides. Cofactor-related participant features remain part of the reaction representation; the separate annotation loss in Section A.6 is used only in the biological ablation.

E EFFICIENT EVALUATION OF THE BIOLOGICAL ABLATION

This section concerns only the auxiliary-loss ablation in Section A.6, not the proposed training objective. For one category, define the weighted sums

$$A_c = \sum_{i \in I_c} w_{ic}, \quad Q_c = \sum_{i \in I_c} w_{ic}^2, \quad \mathbf{s}_c = \sum_{i \in I_c} w_{ic} \hat{\mathbf{u}}_i, \quad (59)$$

and the diagonal correction $V_c = \sum_{i \in I_c} w_{ic}^2 \|\hat{\mathbf{u}}_i\|_2^2$. The ordered off-diagonal confidence mass is $T_c = A_c^2 - Q_c$. Expanding the dot product of the weighted sum gives

$$\sum_{\substack{i,j \in I_c \\ i \neq j}} w_{ic} w_{jc} (1 - \hat{\mathbf{u}}_i^\top \hat{\mathbf{u}}_j) = T_c - \|\mathbf{s}_c\|_2^2 + V_c. \quad (60)$$

Retaining V_c makes the expression exact even when numerical normalization does not return exactly unit norm. For the background, define

$$A_c^{\text{out}} = \sum_{j \in B_c} \bar{w}_j^f, \quad \mathbf{s}_c^{\text{out}} = \sum_{j \in B_c} \bar{w}_j^f \hat{\mathbf{u}}_j. \quad (61)$$

Then $U_c = A_c A_c^{\text{out}}$ and

$$\sum_{i \in I_c} \sum_{j \in B_c} w_{ic} \bar{w}_j^f (1 - \hat{\mathbf{u}}_i^\top \hat{\mathbf{u}}_j) = U_c - \mathbf{s}_c^\top \mathbf{s}_c^{\text{out}}. \quad (62)$$

Background sums are obtained from the total family-annotated sum after subtracting the category members’ contributions using their *family* maximum confidence. This is important when a member’s confidence for category c differs from its largest confidence in the family.

The implementation uses indexed reductions to evaluate these identities. If M_f is the number of annotated memberships in a family and C_f its number of categories, their additional time is proportional to $(N_X + M_f + C_f) d_{\text{bio}}$, with $d_{\text{bio}} = 512$; it does not require a dense $N_X \times N_X$ biological distance matrix. This saving does not remove the separate $N_R \times N_E$ matrix used by the full-graph association loss.

F WHY THE TWO ASSOCIATION LOSSES DIFFER

For a single phase-1 positive, write $D_{ap} = \exp(s_{ap}^{(1)}) + \sum_{n \in N_B(a)} \exp(s_{an}^{(1)})$. Its gradients are

$$\frac{\partial \ell_{\text{dec}}(a, p)}{\partial s_{ap}^{(1)}} = \frac{\exp(s_{ap}^{(1)})}{D_{ap}} - 1, \quad (63)$$

$$\frac{\partial \ell_{\text{dec}}(a, p)}{\partial s_{an}^{(1)}} = \frac{\exp(s_{an}^{(1)})}{D_{ap}}, \quad n \in N_B(a). \quad (64)$$

Other known positives are absent from this positive’s denominator and have zero derivative in this individual term. They receive their own positive terms. Averaging these terms within an anchor ensures that adding more recorded positives does not simply multiply its loss weight.

Phase 2 instead fits a categorical distribution with target probability $1/d_a$ for each of an anchor’s d_a positives. Its per-logit derivative is the predicted probability minus the target probability. Its minimum cross-entropy for unconstrained logits approaches $\log d_a$, the entropy of the uniform-positive target, rather than zero when $d_a > 1$. Consequently, numerical loss values from the two phases are not directly comparable. Both losses use unobserved associations as contrastive alternatives; neither resolves all missing-positive annotation.

G ABLATION ANNOTATION PROVENANCE AND BENCHMARK-SPECIFIC INPUTS

The biological ablation builds categories on the training endpoints of each target. These annotations are not required to train the proposed method. For EnzymeMap, EC assignments and atom mapping are read from the native EnzymeMap cache only at indices recorded in the official training associations. Mechanism labels are derived from the native mapped reactants and products. The native quality field must be at least 0.5, and the retained confidence is capped at one. Cofactor presence is recognized from the training reaction participants and assigned confidence 0.4.

For ReactZyme, native EC annotations are restricted to training protein sequences. Training reaction EC annotations use reaction-only Rhea matching and exact canonical-chemistry lookup. Coarse

bond-change categories use the cached mapping of each training reaction, retain mapping confidence at least 0.5, and multiply this confidence by the reaction-matching confidence. Cofactor categories use the permitted training participant sets. These reaction-only annotation resources provide information beyond the raw released participant-set input; they are auxiliary training supervision, not an inference-time reconstruction of missing reaction direction.

For every training edge, the reaction’s observed categories are added to its associated protein, retaining the maximum confidence when a membership is supported more than once. No held-out reaction–enzyme association is used for this propagation. Unknown or absent categories are omitted. In the current EnzymeMap annotation vocabulary, the observed coarse mechanism groups are phosphate transfer, redox/carbonyl interconversion, stereochemical rearrangement, and sulfur/thiol chemistry. These broad labels should not be described as a detailed model of catalytic mechanisms.

The target training graph and dictionary sizes are given in Table 4. Duplicate association rows are removed when building the compact phase-2 graph. Training endpoints can remain eligible candidates under a benchmark’s candidate-pool definition; the method does not silently remove them.

Target	Training rows	Unique phase-2 edges	Reaction anchors
ReactZyme Reaction-Sim	147,393	147,393	6,977
ReactZyme Enzyme-Sim	152,748	152,748	7,386
ReactZyme Time	149,554	149,554	7,281
EnzymeMap	34,427	34,180	12,603

Table 4: Target-specific training inputs for the implemented recipe. Each row has independently fitted retrieval weights, phase-2 heads, and a dictionary.

H OPTIMIZATION AND INFERENCE PROCEDURE

Setting	Value
Phase-1 seed; association batch size	42; 512
Retained phase-1 snapshot	End of epoch 10
Phase-1 optimizer	AdamW, lr = 10^{-4} , weight decay 10^{-2}
Phase-1 temperature	Fixed $\tau_1 = 0.2$ (inverse 5)
Phase-1 directional weights	1/2 for each direction
Unknown-negative weight	1
Entropy and diversity coefficients	0.01 and 0.01
Protein attention uniform mixture	$\eta = 0.05$
Gate uniform mixture	$\zeta = 0.05$
Phase-2 optimizer	AdamW, lr = 10^{-4} , weight decay 10^{-3}
Phase-2 updates; seed	100 full-graph updates; 42
Phase-2 temperature; identity coefficient	$\tau_2 = 0.2$; 10
Phase-2 residual training coefficient	$\kappa = 0.2$
Phase-2 gradient clipping	Global gradient norm 1
Added biological-label supervision	None in either training stage
Inference protein fusion multiplier	$\nu = 3$
Inference residual multiplier	$c = 0.5$
Dictionary score coefficient	$\alpha = 0.4$
Protein dictionary neighbors	32
Dictionary kernel temperatures	$\tau_E = \tau_R = 0.03$
Reaction dictionary truncation	None; all training reaction anchors

Training and inference procedure.

1. Construct the target training association graph and the frozen feature caches. Initialize the trainable retrieval components for that target and load the frozen pretrained feature resources and SLEEC scorer.

2. Train the phase-1 encoders with Eq. (34). For each batch, deduplicate endpoint IDs and recover all known in-batch training positives. Retain the epoch-10 state.
3. Freeze F3. Export native training embeddings using its learned scale s , without the factor-three inference adjustment. Build the dictionary centers, training protein bank, training reaction bank, and association map.
4. Initialize the residual heads at identity. For each of 100 updates, encode every training endpoint, evaluate the full bidirectional cross-entropy and identity penalty in Eq. (40). Clip gradients and update only the residual heads.
5. At inference, obtain the protein’s global and fused branches and use νs in Eq. (51). Apply the trained heads with residual coefficient $c_K = 0.1$.
6. Independently compute the enzyme and reaction dictionary coordinates from frozen raw features and the stored training dictionary. Concatenate the weighted blocks, or retain equivalent dense and sparse blocks.
7. Rank candidates by Eq. (54) in the requested direction. No candidate labels, candidate-pool normalization, or biological annotation files are needed for encoding new endpoints.

Checkpoint representation of the fusion adjustment. The EnzymeMap deployment checkpoint already stores the factor-three change to the bounded fusion scalar, so its runtime fusion multiplier is one. The ReactZyme deployment checkpoints store the native scalar and apply three at runtime. These represent the same recipe. Applying three again to the calibrated EnzymeMap checkpoint would implement a different model. Phase-2 training used the native feature exports for every target.

Numerical evaluation. Dictionary feature norms, centers, and nearest-neighbor similarities use FP64 reductions where implemented before storing FP32 features. Neighbor selection resolves cutoff ties by the lower training-anchor index. Equivalent mathematical score decompositions can nevertheless have different floating-point rounding when executed in different orders. The frozen evaluation routines and candidate ordering should therefore be retained, particularly for nearly tied ReactZyme reaction candidates. The dictionary is part of the deployed method, not a post-hoc normalization fitted to the current screening library.

I SCOPE OF THE BIOLOGICAL-SUPERVISION ABLATION

The proposed method omits biological-label supervision. The V4+Bio ablation adds it in the second stage; its gradients update only the residual heads over the complete training graph, while the residue queries and reaction encoder retain their first-stage parameters. Category eligibility and background coverage are therefore defined over training endpoints rather than individual minibatches.

Matched removal and shuffled-label controls show mixed, mostly small benchmark changes. The extra unique Case1 catalyst crosses from rank 26 to 25, with unchanged recovery in the original-entry panel. These results do not establish a consistent generalization benefit, motivating omission of the added loss from the proposed method. Equal family coefficients do not imply equal gradient influence or equal predictive benefit. The 512 neural coordinates have no fixed biochemical names; only the dictionary coordinates have an explicit training-reaction interpretation.

J ABLATION OF THE LEARNED RESIDUE-VIEW COUNT

Motivation and capacity. The hyperparameter K determines how many learned residue summaries are available before view fusion. The global mean preserves broad sequence context, and SLEEC provides a fixed functional-residue prior; the learned queries can adapt their pooling to enzyme–reaction associations. Multiple queries can preserve complementary patterns that a single weighted mean would merge. This is a capacity motivation: neither disjoint residue sets nor named biochemical roles are imposed, and additional views may be redundant.

Each extra view adds a 256-dimensional query, an independent projection, and wider gate input and output layers. At the stated hidden and output widths, the trainable multiview enzyme mod-

ule has $P(K) = 3,159,298 + 395,008K$ parameters: 3.554, 3.949, 4.739, and 6.319 million for $K = 1, 2, 4, 8$, respectively. These counts exclude frozen feature resources and SLEEC, the reaction encoder, and the phase-2 heads. Thus $K = 4$ permits several adaptive summaries at a lower parameter cost than $K = 8$, while keeping the final neural embedding dimension fixed at 512.

Matched protocol. We train $K \in \{1, 2, 4, 8\}$ separately on EnzymeMap and the three ReactZyme splits, giving 16 fresh fits with seed 42. The $K = 4$ reference is retrained within this study. All arms use the proposed association-only objective. Within each target, only K and its induced projection and gate dimensions vary. Frozen inputs, association graph, 1,022-residue input limit, SLEEC/global views, losses, and training and inference settings are held fixed: phase 1 uses 10 epochs and batch 512; phase 2 uses 100 updates; inference uses $(\nu, c, \alpha) = (3, 0.5, 0.4)$ (Appendix H). We report every arm at those fixed checkpoints on the same complete benchmark candidate pools. These fresh-run ablation values are distinct from the historical checkpoints in the primary benchmark comparison.

Table 6: Learned residue-view count on ReactZyme. Entries are query-averaged all-positive test MRR; $R \rightarrow E$ and $E \rightarrow R$ denote the retrieval directions. All rows use seed 42 and fresh target-specific training. Bold marks the largest value in each column, without implying significance.

K	Reaction-Sim		Enzyme-Sim		Time	
	$R \rightarrow E$	$E \rightarrow R$	$R \rightarrow E$	$E \rightarrow R$	$R \rightarrow E$	$E \rightarrow R$
1	0.4000	0.5419	0.6632	0.9652	0.5374	0.8042
2	0.4045	0.5158	0.6672	0.9626	0.5381	0.7975
4	0.3947	0.5410	0.6662	0.9704	0.5383	0.7897
8	0.3926	0.5010	0.6674	0.9613	0.5398	0.7911

Table 7: Learned residue-view count on EnzymeMap. BEDROC values are percentages; EF5 and EF10 are fold enrichments. The full library contains 1,521 queries and 261,907 candidates; exclusion of training identifiers leaves 1,337 queries and 252,113 candidates. Bold marks the largest value within each setting and metric; all rows use seed 42.

K	BEDROC ₈₅	BEDROC ₂₀	EF5	EF10
<i>Full library</i>				
1	52.22	70.78	15.82	8.67
2	57.97	75.61	16.73	8.90
4	57.23	75.54	16.86	8.96
8	50.40	69.05	15.59	8.49
<i>Training-ID-excluded library</i>				
1	47.03	67.09	15.22	8.45
2	53.64	72.61	16.23	8.72
4	52.87	72.63	16.41	8.79
8	44.97	65.15	14.90	8.23

Results and retained default. Tables 6 and 7 show no uniformly best count. On EnzymeMap, $K = 2$ gives the highest BEDROC₈₅ in both settings, whereas $K = 4$ gives the highest EF5 and EF10. Relative to $K = 2$, $K = 4$ loses 0.74 and 0.78 BEDROC₈₅ percentage points in the full and excluded libraries, respectively. On ReactZyme, $K = 4$ improves $E \rightarrow R$ MRR over $K = 2$ on Reaction-Sim and Enzyme-Sim, but reduces Reaction-Sim $R \rightarrow E$ and Time $E \rightarrow R$ MRR. Increasing to $K = 8$ yields small gains in Enzyme-Sim and Time $R \rightarrow E$ MRR, but lower EnzymeMap scores and lower Reaction-Sim MRR in both directions than $K = 4$. We therefore retain the original $K = 4$ default as a practical compromise across screening and bidirectional retrieval, rather than claim that it maximizes every metric.

Scope of the evidence. This is a single-seed sensitivity study, not an estimate of robustness or statistical significance. Increasing K also increases parameter count; the comparison is not parameter-

1080 matched, and a common seed does not ensure identical initialization across different shapes. Since
1081 every arm retains at least one learned view, this experiment does not isolate the benefit of adding
1082 learned views relative to a $K = 0$ model. The interpretation applies to the proposed association-only
1083 training setting; it does not establish an optimal count across random seeds or unrelated downstream
1084 tasks.

1085
1086
1087
1088
1089
1090
1091
1092
1093
1094
1095
1096
1097
1098
1099
1100
1101
1102
1103
1104
1105
1106
1107
1108
1109
1110
1111
1112
1113
1114
1115
1116
1117
1118
1119
1120
1121
1122
1123
1124
1125
1126
1127
1128
1129
1130
1131
1132
1133